

Exercise 2

```
abstract class Document {  
    abstract void open();  
}
```

```
class WordDocument extends Document {  
    void open() {  
        System.out.println("Opening Word Document");  
    }  
}
```

```
class PdfDocument extends Document {  
    void open() {  
        System.out.println("Opening PDF Document");  
    }  
}
```

```
class ExcelDocument extends Document {  
    void open() {  
        System.out.println("Opening Excel Document");  
    }  
}
```

```
abstract class DocumentFactory {  
    abstract Document createDocument();  
}
```

```
class WordFactory extends DocumentFactory {  
    Document createDocument() {  
        return new WordDocument();  
    }  
}
```

```
class PdfFactory extends DocumentFactory {  
    Document createDocument() {  
        return new PdfDocument();  
    }  
}
```

```
class ExcelFactory extends DocumentFactory {  
    Document createDocument() {  
        return new ExcelDocument();  
    }  
}
```

```
public class FactoryMethodTest {  
    public static void main(String[] args) {  
  
        DocumentFactory wordFactory = new WordFactory();  
        Document wordDoc = wordFactory.createDocument();  
        wordDoc.open();  
    }  
}
```

```
DocumentFactory pdfFactory = new PdfFactory();  
Document pdfDoc = pdfFactory.createDocument();  
pdfDoc.open();  
  
DocumentFactory excelFactory = new ExcelFactory();  
Document excelDoc = excelFactory.createDocument();  
excelDoc.open();  
}  
}
```

Output:

Output

```
Opening Word Document  
Opening PDF Document  
Opening Excel Document
```

```
=== Code Execution Successful ===
```